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Department of Electronics and Communication Engineering

A Synopsis on Minor Project-1 [22UECL505]

“Gesture control Robotic Arm”

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1. Introduction

Robots are no longer just machines that respond to buttons or pre-programmed commands; they are becoming smarter and more interactive in the way they communicate with humans. One exciting step in this direction is the use of gesture motion control. Human beings naturally express ideas, instructions, and emotions through gesture such as moving a hand to control . Translating this simple yet powerful movements into robot control creates a bridge between human behaviour and machine response, making interaction and more user friendly.

The gesture motion robot experiment is an effort to explore this bridge. Instead of relying on keyboards, remotes, or voice commands, the robot is designed to recognize hand movements and act accordingly. This brings a more natural way of communication, especially for people who may not be familiar with technical interfaces. For instance, a simple forward motion of the hand can guide the robot to move forward, or a sideways gesture can turn it left or right. Such responsiveness gives the user a sense of direct connection with the machine, almost like controlling it with thought and motion rather than complicated commands.

This experiment is important not only because it's new to control a robot with gestures, but also because it can be used in many different ways. Gesture-based control systems are useful in places where voice commands don't work, like in noisy industries, or where hands-free devices are needed, like during medical surgeries or in dangerous situations. It also makes it possible to help people with disabilities who might find it easier to use simple motions than traditional controllers.

This experiment also helps us learn how sensors, microcontrollers, and programming can work together to make machines "aware" of what people are doing. Students and researchers learn more about robotics, artificial intelligence, computer vision, and human-machine interaction by studying and using gesture motion control.

The gesture motion robot experiment is more than just a technical project; it is a step toward making robots easier for people to use and more human-friendly. It shows the goal of making machines that change to fit us, instead of making us change to fit them. This human-centered approach to robotics is promising for the future, when technology will work with people in ways that are easy and natural.

2. Problem Statement

Robotic arms are essential in automation, education, and assistive technology, but traditional control methods still face several limitations:

- **Complex interfaces:** Many robotic arms require joysticks, remotes, or programming skills, which are not easy for beginners.
- **Accessibility issues:** Physical controllers may be challenging for people with mobility impairments.
- **High cost of advanced systems:** Vision-based industrial solutions are often expensive and not suitable for low-budget or educational use.
- **Limited flexibility:** Predefined control schemes lack adaptability for dynamic, real-time interaction.

Purpose

- The goal of this project is to use MediaPipe's real-time hand gesture recognition to create an inexpensive and user-friendly robotic arm control system. The system seeks to do the following by doing away with physical controllers:
- A camera captures the user's hand movements.
- Google's MediaPipe framework processes the video feed, detecting hand landmarks and recognizing specific gestures.
- The recognized gestures are mapped to robotic arm movements through an Arduino (or ESP32) controller that drives servo motors.
- Make it easier for people to interact with robots by letting them use hand gestures to control the arm naturally.
- Encourage accessibility so that people with restricted mobility or technical know-how can more easily engage with robotic systems.
- By giving students, users, and researchers an affordable platform to investigate computer vision and robotics integration, you can promote education and research.

Relevance

- Given the current state of technology and the growing importance of automatic human-machine interaction, this project is extremely relevant. A robotic arm combined with MediaPipe-based hand gesture recognition provides a number of significant benefits:
- Human-Robot Interaction : Enhances human-machine communication by substituting natural gestures for complex controllers.
- For people with limitations in mobility, assistive technology offers a potential solution that makes it easier for them to operate robotic devices.
- Education & Research: Provides researchers and students with an affordable and useful platform to investigate the integration of robotics, computer vision, and machine learning.
- Automation & Industry: In settings where traditional controllers are impractical or hands-free operation is preferred, gesture-controlled robotic systems can increase safety and efficiency.

Literature Survey

- In the field of human-computer interaction, gesture recognition has grown in importance. Conventional systems recorded hand motion using sensors like flex sensors, accelerometers, or gloves. Although these techniques are accurate, they frequently call for extra hardware, which raises the system's cost and reduces its usability.
- With advancements in computer vision, camera-based gesture recognition has gained popularity because it eliminates the need for wearable devices. Frameworks like MediaPipe, developed by Google, allow real-time detection of 21 hand landmarks using a simple camera feed. This makes it possible to recognize gestures reliably even with inexpensive hardware.

- A number of researchers have shown how gesture recognition can be used to control robots, drones, and smart appliances. Specifically, research is being done on gesture-controlled robotic arms for assistive technology and education.
- However, issues like background noise, changing lighting, and misclassified gestures were common in earlier research. In order to overcome this, modern techniques employ lightweight machine learning models and landmark-based recognition, which improves the dependability and affordability of real-time robotic control.
- Thus, combining MediaPipe gesture recognition with a robotic arm offers a viable solution that is affordable, simple to use, and versatile for a range of uses.

Areas of Application

The proposed gesture-controlled robotic arm can be applied in multiple domains:

- **Assistive Technology:** Helping individuals with physical disabilities control robotic aids, wheelchairs, or prosthetics more naturally.
- **Industrial Automation:** Enabling hands-free control of robotic manipulators in manufacturing, assembly, or hazardous environments.
- **Healthcare:** Assisting surgeons or medical staff with sterile, touch-free interaction during operations or in labs.
- **Education and Research:** Serving as a low-cost platform for students and researchers to explore robotics, computer vision, and AI integration.
- **Smart Homes:** Allowing users to control household robotic systems and appliances through intuitive hand gestures.
- **Defense and Disaster Response:** Offering remote gesture-based control of robotic arms for bomb disposal, rescue operations, or handling dangerous materials.
- **Entertainment and Gaming:** Providing immersive gesture-based interactions for robotic toys, VR/AR applications, and interactive exhibits.

Benefits

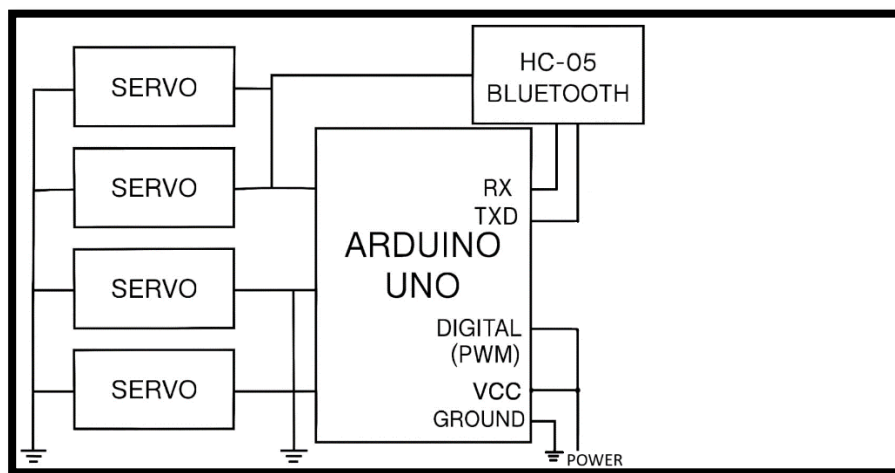
The proposed system offers several key advantages:

- **Intuitive Control:** Users can operate the robotic arm naturally using simple hand gestures, without requiring complex controllers or programming knowledge.
- **Cost-Effectiveness:** The use of a basic camera and open-source MediaPipe framework reduces hardware and software costs compared to specialized sensors or industrial solutions.
- **Accessibility:** Provides an alternative interface for individuals with physical limitations, enhancing inclusivity in human–robot interaction.
- **Real-Time Operation:** MediaPipe enables fast gesture recognition, ensuring smooth and responsive control of the robotic arm.
- **Offline Functionality:** Since MediaPipe can run locally, the system does not rely on constant internet connectivity, making it more reliable in different environments.
- **Flexibility & Customization:** Gesture sets can be easily modified or expanded, allowing adaptation for specific tasks or user preferences.
- **Educational Value:** Serves as a practical tool for students, hobbyists, and researchers to learn about computer vision, AI, and robotics integration.

3. Objectives

- The main goal of this project is to use MediaPipe to design and implement a robotic arm that is controlled by real-time hand gesture recognition.
- By substituting camera-based gesture input for physical controllers, you can create a natural and simple control system.
- Enable hands-free operation to improve accessibility for users, including those with limited mobility.
- Provide an affordable solution that can be used for research and educational purposes.
- Reduce dependency on internet connectivity by running MediaPipe locally to ensure offline operation.
- Make control adaptable and customizable so that new gestures can be assigned to various robotic arm motions.

Block diagram



Features

- **Gesture-Based Control:** Operates using hand gestures detected by a camera through MediaPipe, eliminating the need for physical controllers.
- **Real-Time Processing:** Provides smooth and responsive control of robotic arm movements with minimal delay.
- **Multi-Servo Operation:** Controls multiple servo motors (X, Y, Z, and claw/gripper) for flexible arm movement.
- **Bluetooth Connectivity:** Enables wireless communication between the processing unit (PC/Raspberry Pi) and Arduino/ESP32.
- **Offline Functionality:** Runs locally without the need for constant internet connectivity.
- **Cost-Effective Design:** Uses affordable components (Arduino/ESP32, servos, camera, and Bluetooth module).
- **Customizable Gestures:** Gesture sets can be redefined or expanded for different tasks.

- **Cross-Platform Implementation:** Compatible with PCs, Raspberry Pi, or microcontrollers supporting MediaPipe integration.
- **Educational & Research Friendly:** Provides a practical platform for learning robotics, AI, and human–machine interaction.

Working Principle

- The system works on the concept of vision-based hand gesture recognition integrated with robotic arm control. Its operation can be explained in the following stages:
- **Gesture Capture:**
A camera continuously captures the user's hand movements in real time.
- **Hand Landmark Detection (MediaPipe):**
 - The video feed is processed by MediaPipe Hands, which detects the palm and tracks 21 key landmarks of the hand.
 - These landmarks (x, y, z coordinates) are extracted for further processing.
- **Gesture Recognition:**
 - Specific patterns of hand landmarks are compared against predefined gesture models.
 - Each gesture is assigned to a particular robotic arm action (e.g., open hand → arm up, fist → arm down, point → claw movement).
- **Command Generation:**
 - Once a gesture is recognized, a corresponding control signal is generated.
 - The control signal is sent to the Arduino UNO (or ESP32) through Bluetooth (HC-05 module) or serial communication.
- **Servo Motor Actuation:**
 - The Arduino interprets the received command and drives the servo motors (X, Y, Z, and claw) accordingly.
 - Each servo corresponds to a specific joint or movement of the robotic arm.
- **Robotic Arm Movement:**
 - The robotic arm replicates the user's gesture in real-time.
 - Continuous video feedback ensures smooth interaction between gesture recognition and arm control.

Design Methodology

1. Gesture Recognition via MediaPipe

The project employs Google's MediaPipe framework for hand tracking. MediaPipe provides real-time hand landmark detection, enabling the system to interpret hand gestures through webcam input. This approach minimizes hardware requirements and leverages efficient computer vision techniques.

2. Arduino-Based Robotic Control

An Arduino microcontroller interfaces with the MeArm robotic arm, controlling four servo motors. The system translates recognized gestures into servo movements, allowing for intuitive control of the arm's movements. The servo positions are calibrated to match the physical range of the MeArm's joints.

3. Wireless Communication

The system utilizes a Bluetooth module for wireless communication between the Python application (running on a PC or Raspberry Pi) and the Arduino. This setup eliminates the need for physical connections, enhancing mobility and flexibility in operation.

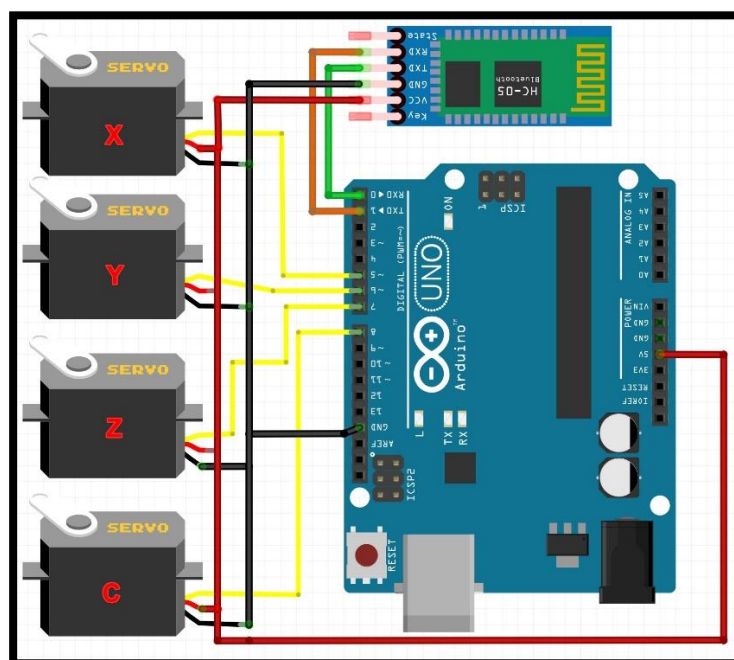
4. User-Centric Calibration

To accommodate variations in user hand sizes and MeArm configurations, the system allows for calibration of servo movement ranges. Users can adjust parameters such as minimum and maximum angles for each servo, ensuring personalized and accurate control.

5. Real-Time Interaction

The design prioritizes real-time responsiveness, enabling immediate translation of hand gestures into robotic movements. This is crucial for applications requiring precise and timely interactions.

Schematic and Circuit diagrams



Algorithms

1. Hand Detection Algorithm

Purpose: Detect hand and track landmarks using a webcam.

Steps:

1. Initialize MediaPipe hand tracking.
2. Capture video frame from webcam.
3. Process frame need to detect hand landmarks.
4. Store landmark coordinates for further processing.

2. Gesture Recognition Algorithm

Purpose: Identify gestures like open palm or fist to control the claw.

Steps:

1. Calculate distances between fingertips and palm base.
2. Compare distances to threshold values.
3. Determine gesture:

Fist → close claw

Open palm → open claw

3. Coordinate Mapping Algorithm

Purpose: Convert hand position in camera coordinates to servo angles for MeArm.

Steps:

1. Read hand coordinates (x, y, z).
2. Apply linear interpolation to map coordinates to servo angles.
3. Ensure servo angles are within mechanical limits.

4. Servo Control Algorithm

Purpose: Send commands to Arduino to move the MeArm servos.

Steps:

1. Open serial communication with Arduino.
2. Send computed servo angles.
3. Arduino receives angles and moves each servo to the corresponding position.

5. Real-Time Loop Algorithm

Purpose: Continuously detect gestures and control MeArm.

Steps:

1. Capture frame from webcam.
2. Detect hand landmarks.
3. Recognize gesture.
4. Map coordinates to servo angles.
5. Send angles to Arduino.
6. Repeat in a loop for real-time response.

List of Hardware and Software components and tools required

Hardware Components

- MeArm Robotic Arm
- Arduino Board (e.g., Arduino Uno)
- Servo Motors (4)
- Webcam or Android phone
- Bluetooth Module (optional)
- Power Supply / Batteries
- Cables & Connectors
- Mounting Hardware (screws, brackets, base)

Software Components

- Python 3
- MediaPipe Library
- OpenCV Library
- PySerial Library
- Arduino IDE
- DroidCam (optional)

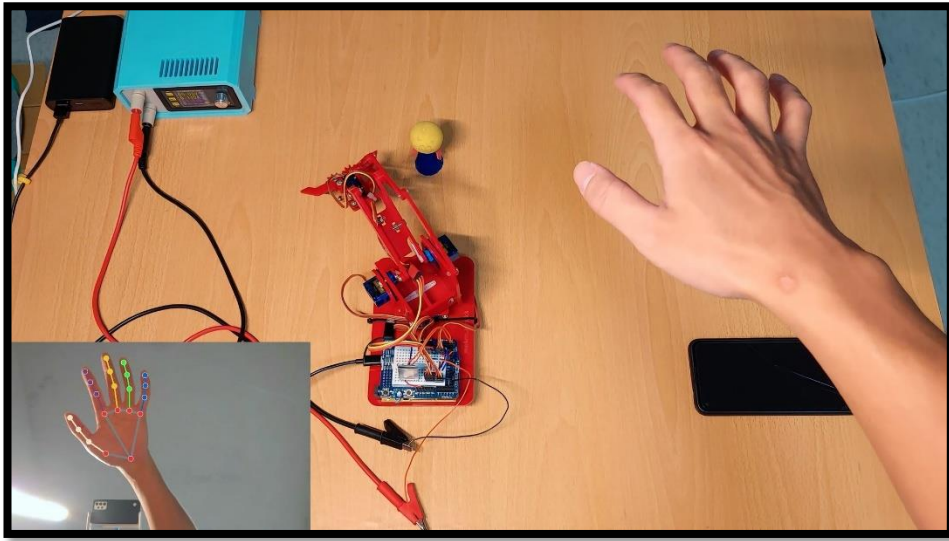
Tools

- Screwdriver Set
- Computer / Laptop
- USB Cable
- Multimeter (optional)
- Soldering Kit (optional)

Expected Results

- Accurate Hand Gesture Detection
 - The system should reliably recognize gestures such as open palm and fist in real-time.
 - Hand positions are tracked accurately with minimal lag.
- Precise Robotic Arm Movement
 - The MeArm robotic arm moves smoothly according to hand movements.
 - Servo motors follow the mapped coordinates of the hand in X, Y, Z directions.
- Claw Control
 - The claw opens when an open palm gesture is detected.
 - The claw closes when a fist gesture is detected.
- Real-Time Response
 - Minimal delay between gesture detection and robotic arm movement.
 - Continuous operation without jitter or misalignment.
- Customizable Calibration
 - Users can adjust servo angle ranges and gesture thresholds for better accuracy.
- Interactive and Educational Application
 - Demonstrates the integration of computer vision, gesture recognition, and robotic control.
 - Can be used for simple tasks like picking and placing small objects.

Expected result outcome :



References:

- Indonesian Journal of Electrical Engineering and Informatics (IJEI) Vol. 11, No. 3, September 2023, pp. 727~737 ISSN: 2089-3272, DOI: 10.52549/ijeei.v11i3.4491
- **Publication:** IET Conference Proceedings Volume 2024, Issue 18
<https://doi.org/10.1049/icp.2024.3947>